



Yifan Zhang

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Location: San Diego, CA **Cybersecurity Center @ SDSU**

I am recruiting motivated PhD students in systems security and privacy.

Research Directions

My research focuses on system security and privacy, with an emphasis on identifying new attack vectors, building automated binary-analysis tools, and conducting large-scale real world measurement. Current on going projects include:

- **Software Supply Chain Security & Privacy**
- **Mobile & IoT Systems Security**
- **Cybercrime**

I emphasize **analysis methods for systems security**; application scenarios are not restricted. Planned explorations include:

- **Blockchain L2** systems' usability, security, and privacy-preserving features.
- **AI/LLM-powered** techniques to facilitate program/system analysis

How We Work

We combine reverse engineering with static & dynamic program analysis and leverage formal verification and AI where applicable to detect, measure, and mitigate novel security & privacy vulnerabilities.

What I Look For in a PhD Student

- **Deep curiosity** about real systems and willingness to dissect complex software stacks.
- **Strong systems skills**: confident with Linux, scripting, and debugging; comfortable reading specs and source.
- **Program analysis**: experience/interest in reverse engineering; static/dynamic analysis (e.g., program slicing, taint/flow analysis, fuzzing), or *formal methods*.

- **Solid coding ability**; able to build research prototypes and measurement pipelines.
- **Team fit**: collaborative, respectful, and proactive; excited to publish in top security venues.

About SDSU

San Diego State University (SDSU) is a major public research university in Southern California and an **R1 university** (Very High Research Activity).

About San Diego

San Diego offers year-round mild weather, beautiful coastline, and a dynamic innovation ecosystem spanning software, telecommunications, biotech/life sciences, defense, and clean-tech. The region's blend of industry, research institutions, and quality of life makes it a compelling place to conduct systems and security research and to live during graduate study.

Interested in Joining?

If you are excited about **systems security and privacy**, and believe you may be a good fit, please prepare:

- A brief statement describing your research interests and any relevant past projects.
- A CV or résumé with links to code or artifacts.
- A link to any open-source contributions or technical write-ups you are proud of.

Email subject template:

[SDSU Security PhD] Your Name --- [Brief Research Interest]